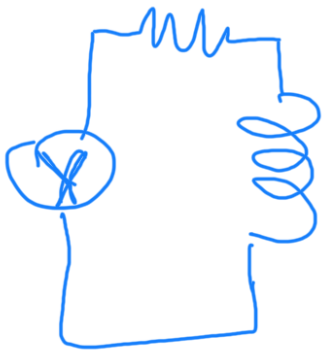
$$\frac{x(t)}{L} = i\frac{R}{L} + \frac{di}{dt} \rightarrow y(t)$$



$$\Rightarrow \left(e^{Rt/L}\right)\left[iR/L + di/dt\right] = \frac{x(t)}{L} e^{Rt/L}$$

$$\Rightarrow \frac{d}{dt}\left[i e^{Rt/L}\right] = \frac{x(t)}{L} e^{Rt/L}$$

$$\Rightarrow i \cdot e^{Rt/L} = \int_{-\infty}^{t} \frac{x(t)}{L} e^{Rt/L} dt$$

$$\Rightarrow L i(t) = e^{-Rt/L} \int_{-\infty}^{t} x(t)\, e^{Rt/L} dt$$

$$\Rightarrow L\frac{di(t)}{dt} = e^{-Rt/L}\, x(t)\, e^{Rt/L} - \frac{R}{L}\left(\int_{-\infty}^{t} x(t) e^{Rt/L} dt\right) e^{-Rt/L}$$

$$\Rightarrow L\, di/dt = x(t) - R/L\, e^{-Rt/L} \int_{-\infty}^{t} \left(x(t) e^{Rt/L} dt\right)$$

$$\Rightarrow -h(t) = \delta(t) - R/L\, e^{-Rt/L} \int_{-\infty}^{t} \delta(t) e^{Rt/L} dt$$

$$\Rightarrow \boxed{h(t) = R/L\, e^{-Rt/L}\, u(t)}$$